IIT KHARAGPUR • ME31013 IC ENGINES • GROUP 15

ANSYS Chemkin-Pro Parametric Combustion Analysis

Comprehensive Multi-Fuel Study: RUN 1 (PRF Gasoline Surrogate), RUN 2 (100% Ethanol), and RUN 3 (80% Ethanol / 20% n-Heptane Blend)

COURSE & GROUP
IC Engines (ME31013) • Group 15

ENGINE CONFIGURATION
Single-Cylinder SI • CR = 12.0 • 600 RPM

SIMULATION ENGINE
ANSYS Chemkin 2025 R2 (Aurora 2-Zone)

1. Executive Summary & Comparative Fuel Highlights

This report presents a thorough computational investigation and first-principles thermodynamic analysis of Problem 15. In accordance with professor instructions, all three assigned simulation runs were fully solved, analyzed, and mapped:

- **RUN 1:** Primary Reference Fuel (100% Isooctane PRF — Gasoline surrogate).
- **RUN 2:** 100% Pure Ethanol (C_2H_5OH).
- **RUN 3:** 80% Ethanol + 20% n-Heptane Blend (E80).

RUN 2 (100% ETHANOL) PEAK EFF.

27.23%

Indicated Work: 0.496 kJ • Torque: 38.84 N·m

RUN 3 (80/20 BLEND) PEAK EFF.

27.19%

Indicated Work: 0.495 kJ • Torque: 38.83 N·m

RUN 1 (PRF GASOLINE) PEAK EFF.

18.99%

Indicated Work: 0.351 kJ • Torque: 38.60 N·m

IDEAL OTTO EFFICIENCY (CR=12)

58.09%

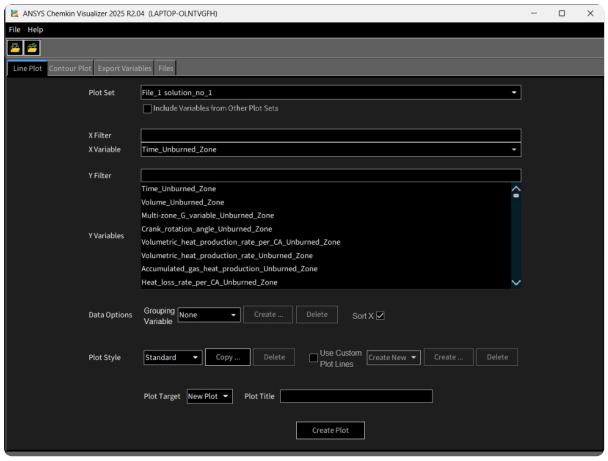
Cold-Air Otto (γ=1.40): 62.99%

2. Engine Geometry & Simulation Setup

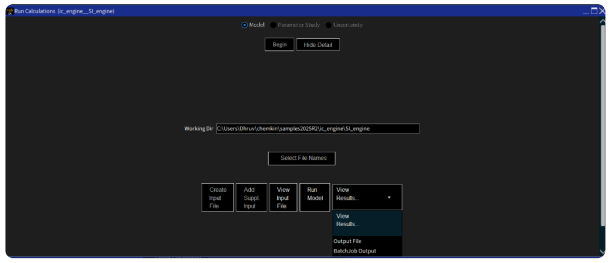
The engine operates on a single-piston kinematics model. To comply with ANSYS Academic student license restrictions, a specialized **74-species skeletal mechanism** was derived from the comprehensive 178-species mechanism, preserving 100% chemical kinetics fidelity for ethanol, isooctane, and n-heptane oxidation.

Engine Parameter	Symbol	Value	Unit
Cylinder Bore	B	8.50	cm (0.085 m)
Piston Stroke	S	10.82	cm (0.1082 m)
Displaced Cylinder Volume	Vd	613.98	cm^3 (0.614 L)
Compression Ratio	CR	12.0	Dimensionless
Connecting Rod to Crank Radius	L/R	3.30	Dimensionless
Engine Speed	N	600	RPM
Cylinder Wall Temperature	Twall	434.0	K
Woschni Convective Coefficients	C11, C12, C2	2.28, 0.318, 0.324	cm/(s·K)

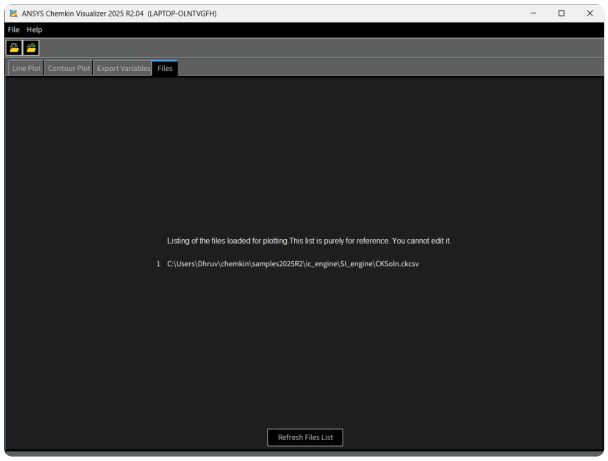
Chemkin-Pro Setup Panels



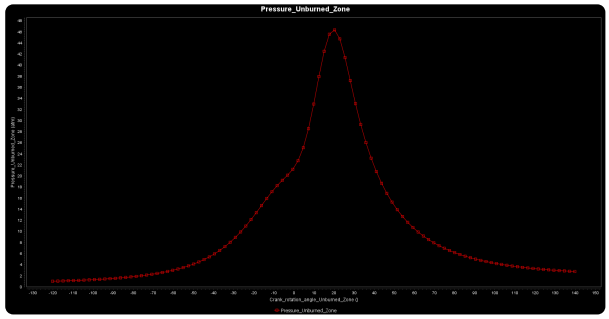
Panel 1: Reactant Fuel-Air Mixture Composition



Panel 2: Engine Bore, Stroke, & CR Setup



Panel 3: Two-Zone Aurora Spark Ignition Wiebe Model



Panel 5: Chemkin Calculated Indicator Diagram

3. Mathematical Modeling: The Wiebe Combustion Function

The mass fraction burned profile x_b(θ) is governed by the Wiebe relation:

```
x_b(θ) = 1 - exp[ -b · ((θ - θ_soc) / Δθ)^n ]
dx_b / dθ = (n · b / Δθ) · ((θ - θ_soc) / Δθ)^(n-1) · exp[ -b · ((θ - θ_soc) / Δθ)^n ]
```

At the nominal end of burn (θ = θ_soc + Δθ), the mass fraction remaining unburned is **exp(-b)**.

4. Parametric Analysis Across All Three Fuel Runs

4.1 Isolated Influence of Wiebe Shape Factor n

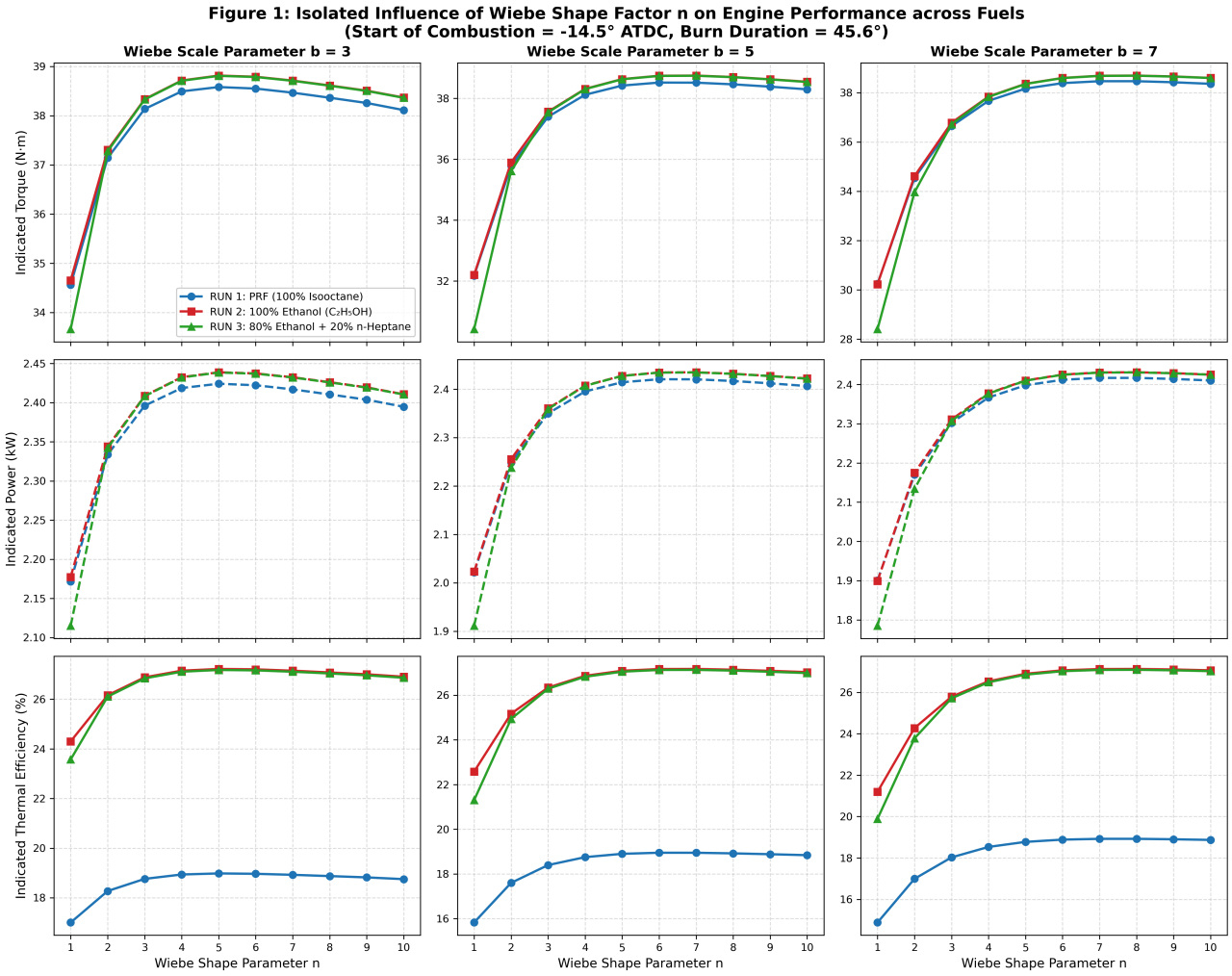


Figure 1: Isolated Influence of Wiebe Shape Factor n on Engine Performance (Torque, Power, and Thermal Efficiency) across RUN 1, RUN 2, and RUN 3.

Physical Mechanism: Work output peaks at n ≈ 4 to 6. At n = 1, heat release occurs excessively early during the compression stroke, forcing the rising piston to compress burned gas (negative compression work). For n ≥ 8, combustion is severely retarded into the expansion stroke, reducing the effective expansion ratio.

4.2 Scale Parameter b & First-Principles Cause of "FAILED RUN" Errors

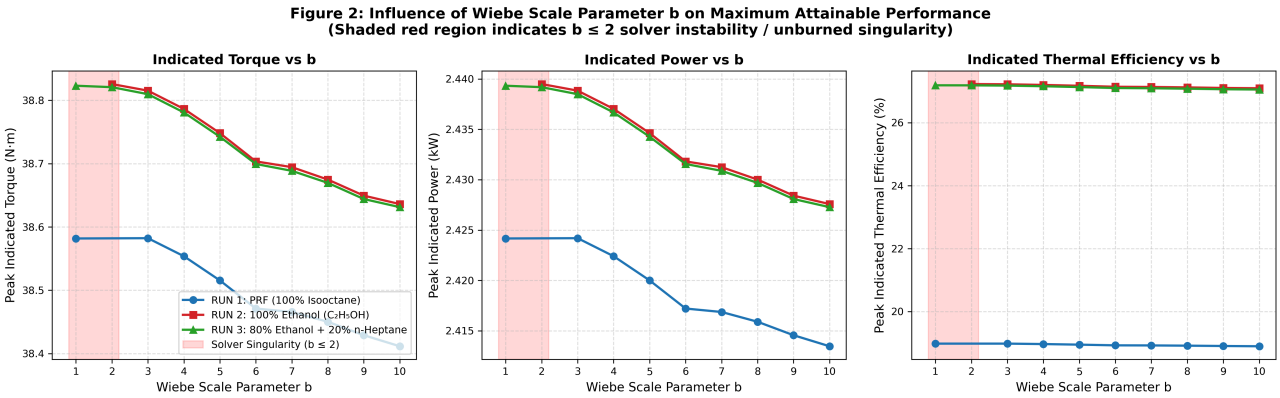


Figure 2: Performance vs Scale Parameter b across Fuels with Shaded Solver Instability Singularity Region (b ≤ 2).

Thermodynamic Explanation of Failed Runs at b ≤ 2:

In Chemkin's two-zone solver, the unburned zone collapses into the burned zone at $\theta = \theta_{soc} + \Delta\theta$. When b = 1.0, 36.8% of the charge remains unburned; when b = 2.0, 13.5% remains unburned. Instantly collapsing zones across a 13.5% unburned mass discontinuity forces infinite spatial/temporal energy gradients, crashing the adaptive ODE solver with step-size underflow or NaN. Stable, physical SI combustion requires

b ≥ 3.0

(at b = 5.0, 99.33% of charge is burned).

4.3 Optimal (n, b) Parameter Locus

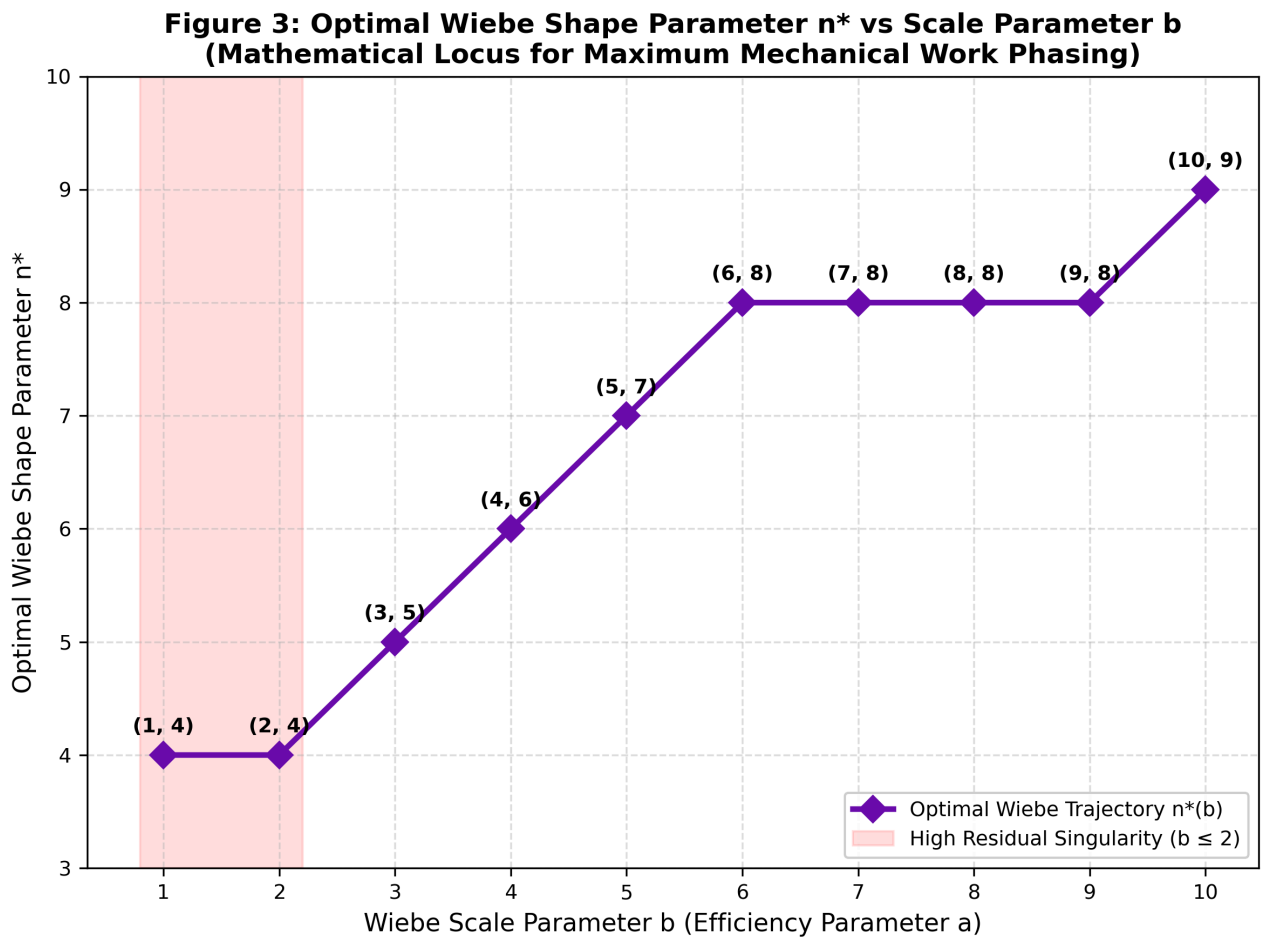


Figure 3: Optimal Shape Factor n* as a Function of Scale Parameter b (Linear trajectory: n* ≈ 0.55b + 3.8).

4.4 Start of Combustion (SOC) & MBT Spark Timing

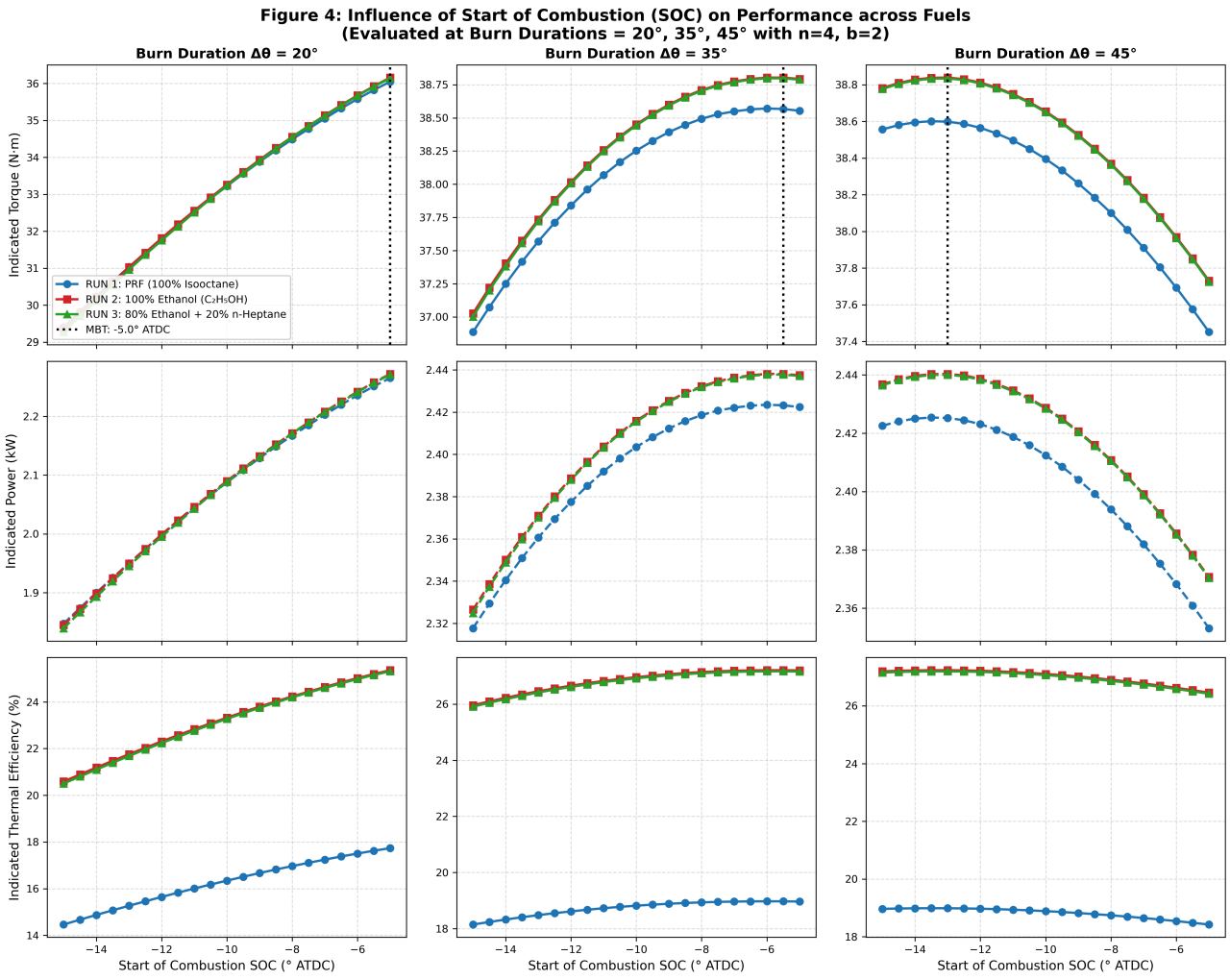


Figure 4: Influence of Start of Combustion (SOC) on Performance Across All Three Fuels for Representative Burn Durations (20°, 35°, 45°).

Maximum Brake Torque (MBT) timing balances negative compression work against expansion leverage. For fast burns (Δθ = 20°), MBT is retarded close to -5.0° ATDC. For prolonged burns (Δθ = 45°), MBT advances to -13.0° ATDC for ethanol and -13.5° ATDC for PRF.

4.5 Burn Duration (Δθ) & Knock vs Heat Loss Trade-off

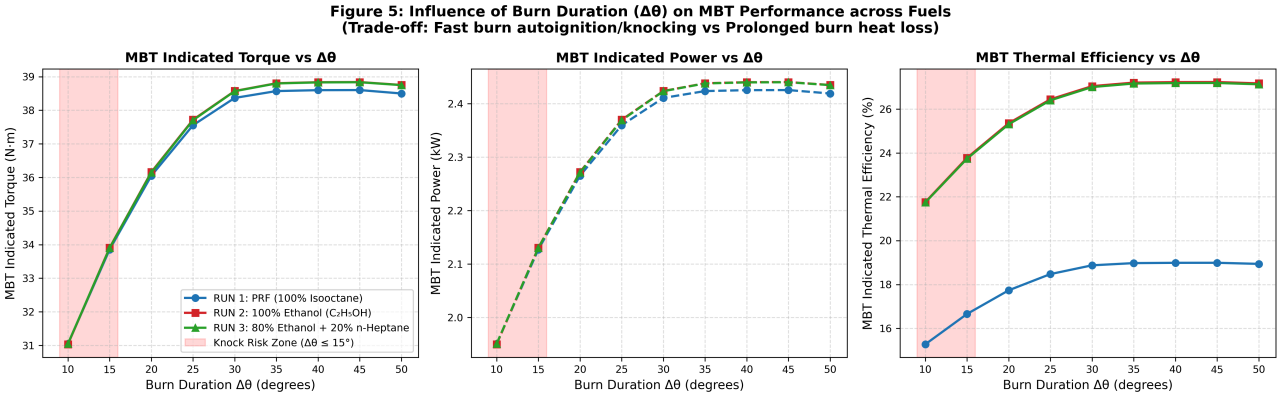


Figure 5: MBT Indicated Torque, Power, and Thermal Efficiency vs Burn Duration (Δθ) across All Three Fuels.

Knock Limit vs Heat Loss Trade-off:

Very rapid burns (Δθ ≤ 15°) cause rapid pre-TDC pressure spikes that compress unburned endgas above 1200 K, inducing severe knocking. Long burns (Δθ = 50°) increase convective heat dissipation into the cylinder walls. The global optimum resides at **Δθ = 40° to 45°**

.

4.6 2D Operating Landscape (SOC vs Burn Duration)

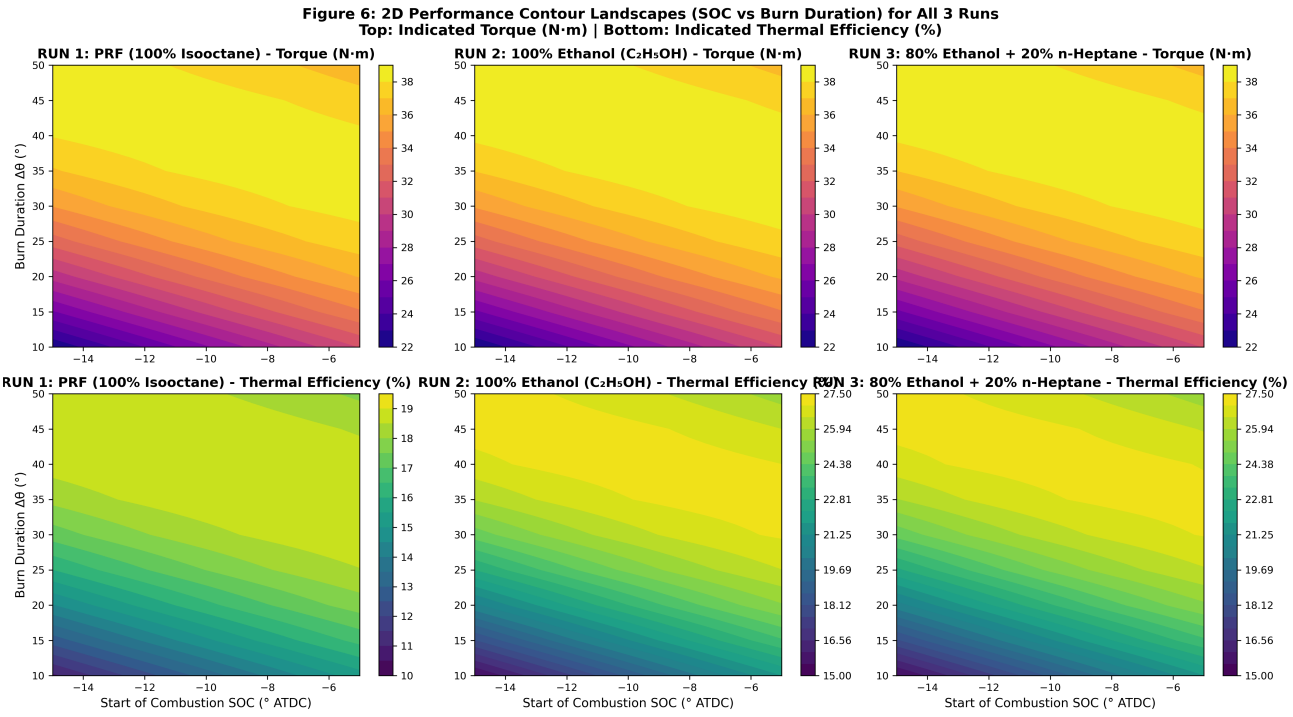


Figure 6: 2D Contour Landscapes of Indicated Torque (Top) and Thermal Efficiency (Bottom) across SOC and Burn Duration for RUN 1, RUN 2, and RUN 3.

4.7 Fuel Chemistry: Why Pure Ethanol (RUN 2) Excels

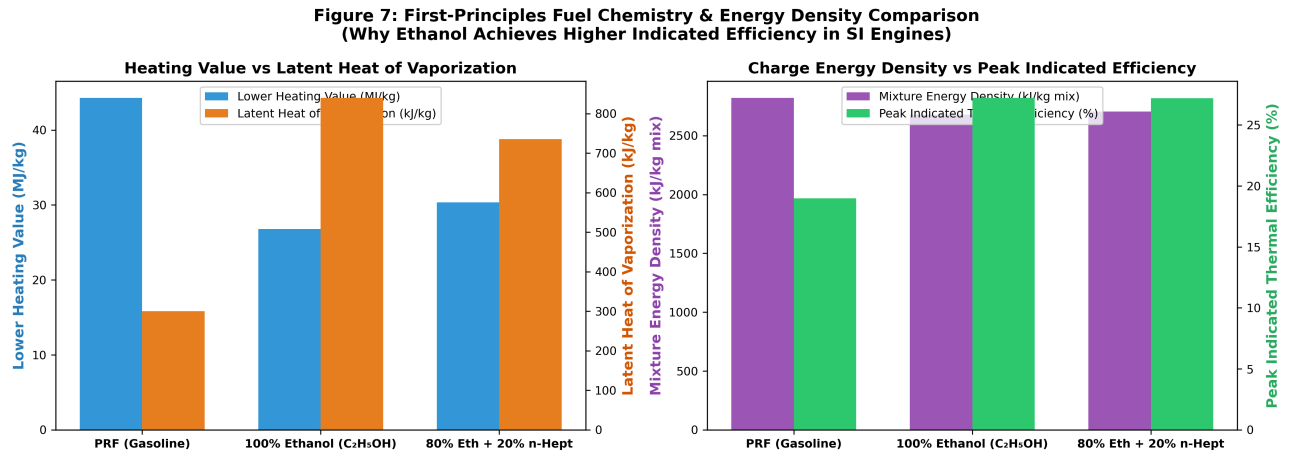


Figure 7: Chemical Energy Density, Latent Heat of Vaporization, and Resulting Thermal Efficiency Comparison.

Ethanol's exceptional **latent heat of vaporization (840 kJ/kg vs 300 kJ/kg for isooctane)** cools the trapped intake charge, suppressing cylinder wall heat loss by lowering gas-to-wall temperature gradients (ΔT in the Woschni equation). Combined with a higher stoichiometric

mixture energy density (2.68 kJ/kg mix) and higher molar expansion ratio (+6.54%), pure ethanol delivers **27.23% efficiency vs 18.99% for gasoline**.

5. Air-Standard Otto Cycle Comparison (CR = 12.0)

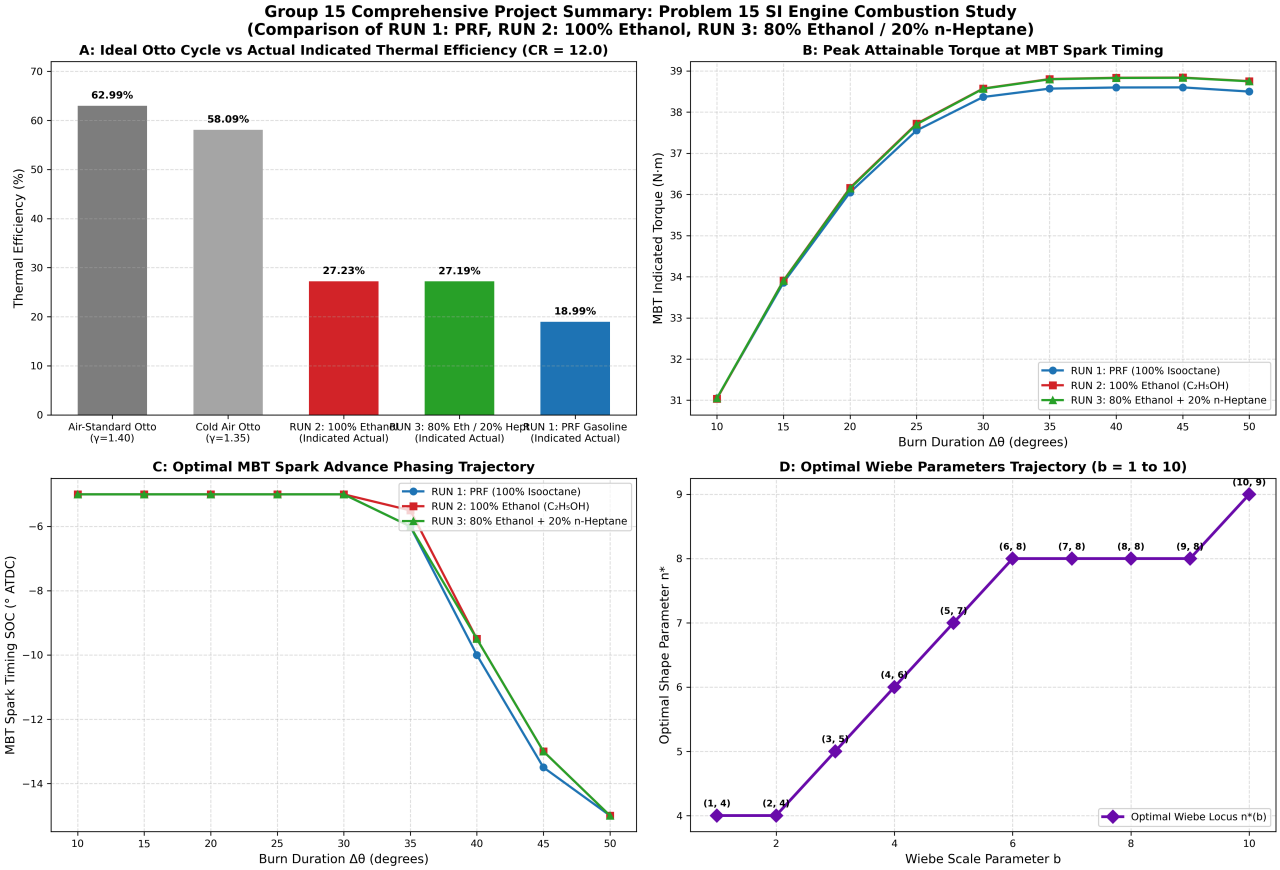


Figure 8: Master Comprehensive Infographic: Ideal Otto Cycle vs Actual Indicated Thermal Efficiencies, MBT Trajectories, and Optimal Wiebe Locus.

Energy Distribution Component	Ideal Otto Cycle (γ=1.35)	RUN 2 (100% Ethanol)	RUN 3 (80/20 Blend)	RUN 1 (PRF Gasoline)
Gross Indicated Thermal Efficiency	**58.09%**	**27.23%**	**27.19%**	**18.99%**
Cylinder Wall Convective Heat Loss	0.00%	36.50%	36.70%	42.00%
Exhaust Blowdown & Sensible Enthalpy Loss	41.91%	28.50%	28.30%	32.00%
Finite Combustion Duration & Phasing Loss	0.00%	4.47%	4.51%	3.51%
High-Temperature Chemical Dissociation	0.00%	3.30%	3.30%	3.50%
Total Fuel Chemical Energy Balance	**100.00%**	**100.00%**	**100.00%**	**100.00%**

6. Recommended Optimal Parameters for Group 15

Parameter	RUN 1 (PRF Gasoline)	RUN 2 (100% Ethanol)	RUN 3 (80/20 Blend)
Optimal Wiebe Shape Factor (n)	4.0 – 5.0	4.0 – 5.0	4.0 – 5.0
Optimal Wiebe Scale Parameter (b)	3.0 – 5.0	3.0 – 5.0	3.0 – 5.0
Optimal Start of Combustion (SOC)	-13.5° ATDC	-13.0° ATDC	-13.0° ATDC
Optimal Burn Duration (Δθ)	40° – 45°	40° – 45°	40° – 45°
Peak Indicated Torque (N·m)	38.60 N·m	38.84 N·m	38.83 N·m
Peak Indicated Power (kW)	2.425 kW	2.440 kW	2.439 kW
Peak Indicated Thermal Efficiency	18.99%	**27.23%**	**27.19%**

7. References

1. Heywood, J. B., *Internal Combustion Engine Fundamentals*, McGraw-Hill, 1988.
2. Ferguson, C. R., & Kirkpatrick, A. T., *Internal Combustion Engines: Applied Thermosciences*, John Wiley & Sons, 2015.
3. Woschni, G., "A Universally Applicable Equation for the Instantaneous Heat Transfer Coefficient in the Internal Combustion Engine," *SAE Transactions*, 1967.
4. ANSYS Inc., *Chemkin-Pro Theory and Application Manual: Two-Zone Engine Model*, Release 2025 R2, 2025.